

Environment:

- Windows 11(year 2026 update)
- Wsl (wsl.exe)
 - PS C:\Users\shad-personal> wsl -v -l
 - WSL version: 2.7.10.0
 - Kernel version: 6.18.33.2-2
 - WSLg version: 1.0.73.2
 - MSRDC version: 1.2.6676
 - Direct3D version: 1.611.1-81528511
 - DXCore version: 10.0.26100.1-240331-1435.ge-release
 - Windows version: 10.0.26200.8457
- Nvidia

```
PS C:\Users\shad-personal> nvidia-smi
```

Sat Aug 1 11:41:56 2026

NVIDIA-SMI 573.22			Driver Version: 573.22			CUDA Version: 12.8		
GPU	Name	Perf	Driver-Model	Bus-Id	Disp.A	Volatile	Uncorr.	ECC
Fan	Temp		Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M.	MIG M.
0	Quadro	P1000	WDDM	00000000:01:00:0	On			N/A
N/A	49C	P8	N/A / 5001W	1001MiB /	4096MiB	3%	Default	N/A

Processes:							
GPU	GI	CI	PID	Type	Process name	GPU Memory	Usage
	ID	ID					
0	N/A	N/A	1400	C+G	...ntrolPanel\SystemSettings.exe		N/A
0	N/A	N/A	3084	C+G	...s\Mozilla Firefox\firefox.exe		N/A
0	N/A	N/A	11008	C+G	...Canon\Quick Menu\CNQMMMAIN.EXE		N/A
0	N/A	N/A	11772	C+G	...yb3d8bbwe\WindowsTerminal.exe		N/A
0	N/A	N/A	12380	C+G	...0.4078.105\msedgewebview2.exe		N/A
0	N/A	N/A	14812	C+G	...2txyewy\CrossDeviceResume.exe		N/A
0	N/A	N/A	14852	C+G	...indows\System32\ShellHost.exe		N/A
0	N/A	N/A	15480	C+G	...4__8wekyb3d8bbwe\ms-teams.exe		N/A
0	N/A	N/A	17096	C+G	...Chrome\Application\chrome.exe		N/A
0	N/A	N/A	19168	C+G	...s\Mozilla Firefox\firefox.exe		N/A
0	N/A	N/A	19296	C+G	...0.4078.105\msedgewebview2.exe		N/A
0	N/A	N/A	20212	C+G	...xyewy\ShellExperienceHost.exe		N/A
0	N/A	N/A	20900	C+G	...cw5n1h2txyewy\SearchHost.exe		N/A
0	N/A	N/A	22728	C+G	...L\wslsettings\wslsettings.exe		N/A
0	N/A	N/A	22768	C+G	C:\Windows\explorer.exe		N/A
0	N/A	N/A	22892	C+G	...Chrome\Application\chrome.exe		N/A
0	N/A	N/A	23676	C+G	...0.4078.105\msedgewebview2.exe		N/A
0	N/A	N/A	23736	C+G	...5n1h2txyewy\TextInputHost.exe		N/A
0	N/A	N/A	25320	C+G	...em32\ApplicationFrameHost.exe		N/A
0	N/A	N/A	25368	C+G	...y\StartMenuExperienceHost.exe		N/A

- Distros under wsl for rkllm toolkit for conversion
 - PS C:\Users\shad-personal> wsl --list
 - Windows Subsystem for Linux Distributions:
 - Ubuntu-24.04 (Default)

Step for deployment of rkllm-tool(We want to utilize nvidia card for better performance)

1. Connect Ubuntu-24.04

```
$> wsl -d Ubuntu-24.04
```

2. Install python and create python virtual environment

```
$> sudo apt update -y
```

```
$> sudo apt install python3-pip python3-venv python3-dev git git-lfs -y
```

```
$> python3 -m venv rkllm_env
```

```
$> cd rkllm_env/
```

```
ubuntu@LAPTOP-CM103FLE:~/rkllm_env$ source bin/activate
```

```
(rkllm_env) ubuntu@LAPTOP-CM103FLE:~/rkllm_env$ bin/python3 --version
Python 3.12.3
```

```
$> bin/python3 -m pip install --upgrade pip    # upgrade pip
```

3. Install pytorch nvidia cuda compliant pytorch pip package

```
$> bin/python3 -m pip install torch torchvision torchaudio --extra-index-url
```

```
https://pytorch.org
```

Note: There is this command “bin/python3 -m pip install torch torchvision torchaudio --index-url <https://pytorch.org>” do not work anymore. So use above command.

Note: If nvidia too old, error encountered

```
"""
```

```
python3 -c "import torch; print(torch.cuda.is_available())"
```

```
/home/ubuntu/rkllm_env/lib/python3.12/site-packages/torch/cuda/__init__.py:188:
```

```
UserWarning: CUDA initialization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and installing a new version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTorch version that has been compiled with your version of the CUDA driver. (Triggered internally at
```

```
/__w/pytorch/pytorch/c10/cuda/CUDAFunctions.cpp:119.)
```

```
  return torch._C._cuda_getDeviceCount() > 0
```

```
False
```

```
"""
```

Update nvidia driver in host by downloading(<https://www.nvidia.com/en-us/drivers/>) and installing as express. Then try below

```
$> bin/python3 -c "import torch; print(torch.cuda.is_available())"
```

```
True
```

—

```
$> bin/python3 -c "import torch; print(torch.cuda.is_available())"
True
```

4. Install rknn-llm toolkit install

```
$> git clone https://github.com/airockchip/rknn-llm.git
```

```
$> cd ~/rknn_llm/rknn-llm/rkllm-toolkit/packages && ls
```

```
requirements.txt          rkllm_toolkit-1.3.0-cp311-cp311-linux_x86_64.whl
requirements_rwkv7.txt    rkllm_toolkit-1.3.0-cp312-cp312-linux_x86_64.whl
rkllm_toolkit-1.3.0-cp310-cp310-linux_x86_64.whl rkllm_toolkit-1.3.0-cp39-cp39-linux_x86_64.whl
```

```
$> export BUILD_CUDA_EXT=0
```

```
$> ~/rkllm_env/bin/python3 -m pip install
```

```
rkllm_toolkit-*-cp312-cp312-linux_x86_64.whl
```

Test rkllm api if no error good to go:

```
$> ~/rkllm_env/bin/python3 -c "from rkllm.api import RKLLM"
```

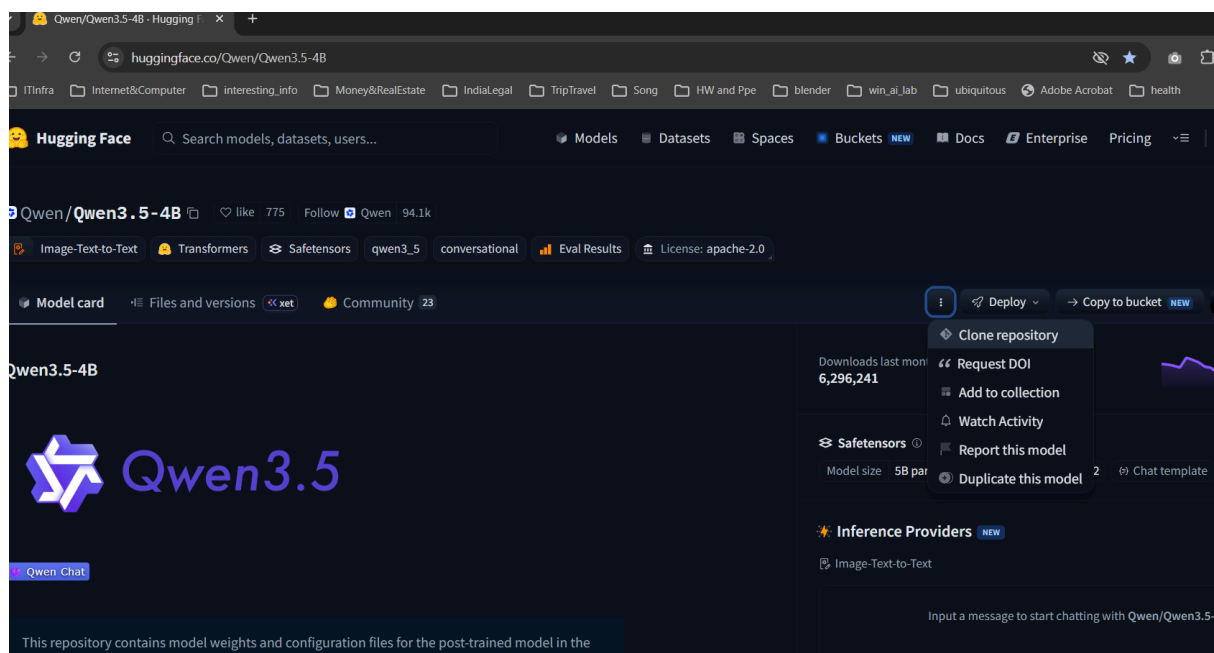
Directive for rkllm to use graphics card

```
$> CUDA_VISIBLE_DEVICES=0 <python command>
```

5. Download Qwen3.5-4b model on host pc and transfer to wsl distros ubuntu

Go to hugging face portal and suggested in screenshot link to copy for qwen3.5-4b through clone repository:

<https://huggingface.co/Qwen/Qwen3.5-4B>



```
Clone this model repository

HTTPS SSH

# Make sure git-xet is installed (https://hf.co/docs/hub/git-xet)
winget install git-xet

git clone https://huggingface.co/Qwen/Qwen3.5-4B

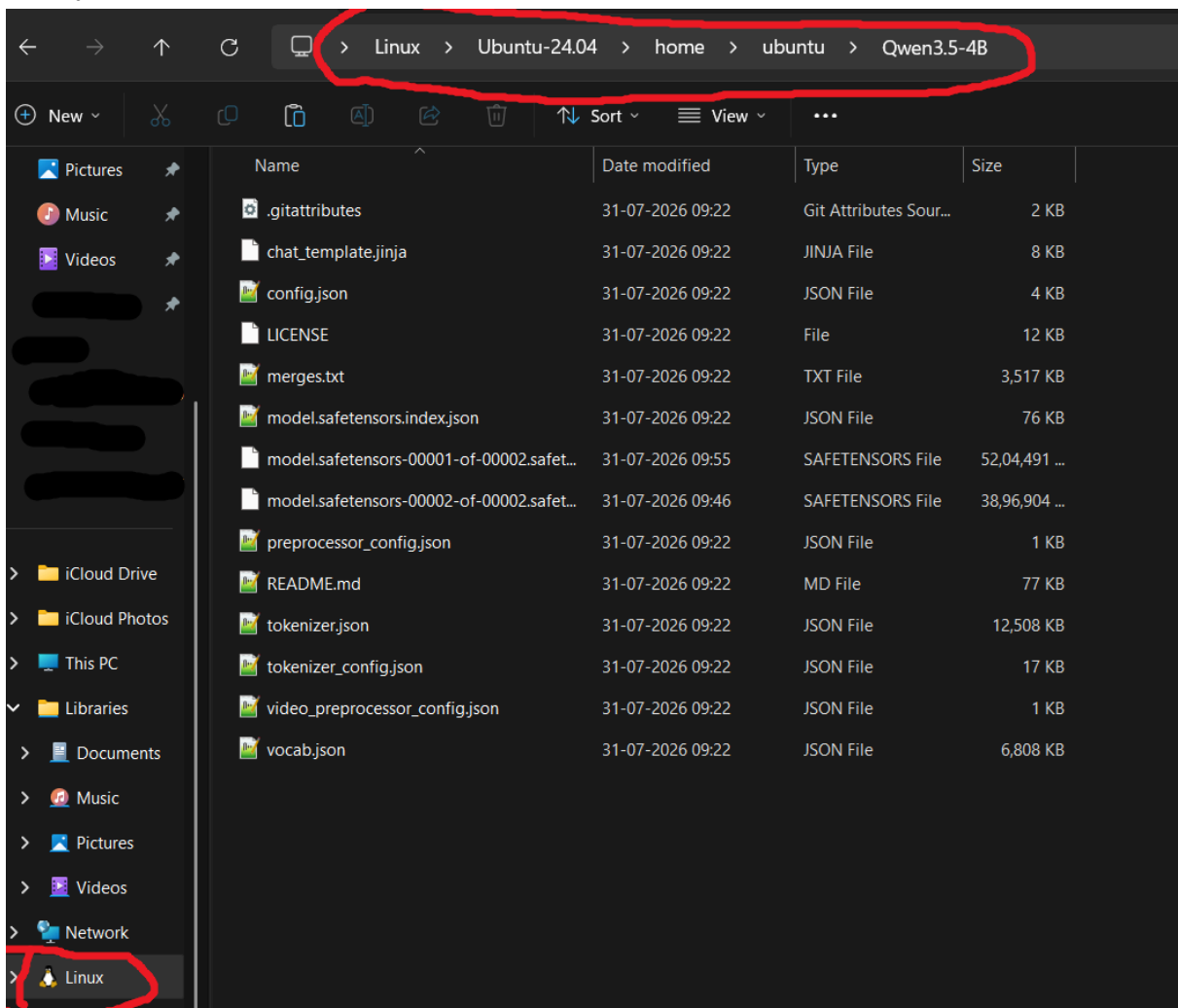
# If you want to clone without large files - just their pointers
GIT_LFS_SKIP_SMUDGE=1 git clone https://huggingface.co/Qwen/Qwen3.5-4B

PowerShell uv uvx

# Make sure the hf CLI is installed
powershell -ExecutionPolicy ByPass -c "irm https://hf.co/cli/install.ps1 | iex"
```

Just git clone qwen in PC from portal. Please refer above portal.

Then you move to wsl:



Note: we installed rkllm-toolkit 1.3.0 . Hence you need to compatible converter script
INFO: rkllm-toolkit version: 1.3.

Compatible converter script:

```
"""
```

```
from rkllm.api import RKLLM
import os
```

```
# Ensure Python knows to use the GPU via CUDA environment arrays
os.environ['CUDA_VISIBLE_DEVICES'] = '0'
```

```
# 1. Initialize the correct RKLLM instance
llm = RKLLM()
```

```
# 2. Load the Hugging Face structure using the exact API name
print("Loading Hugging Face model directory into memory...")
model_path = './Qwen3.5-4B' # Update this to your absolute or relative model path
ret = llm.load_huggingface(model=model_path, device='cuda')
```

```
if ret != 0:
    print("Failed to load the Hugging Face model files!")
    exit(ret)
```

```
# 3. Trigger the math calculation matrices to build and quantize for NPU
print("Building and quantizing the model to w8a8 precision for RK3588...")
ret = llm.build(
    do_quantization=True,
    optimization_level=1,
    quantized_dtype='w8a8',
    quantized_algorithm='normal',
    target_platform='rk3588',
    num_npu_core=3
)
```

```
if ret != 0:
    print("Model optimization compilation build phase failed!")
    exit(ret)
```

```
# 4. Save out the standalone execution artifact file
print("Exporting the final deployment binary file...")
save_path = './qwen3_5_4b.rkllm'
ret = llm.export_rkllm(save_path)
```

```
if ret != 0:
    print("Export phase failed!")
    exit(ret)
```

```
print("Success! Your deployment asset 'qwen3_5_4b.rkllm' is ready for your board.")
```

```
"""
```

Run converter

```
$>CUDA_VISIBLE_DEVICES=0 ~/rkllm_env/bin/python3 qwen_rkllm_converter.py
```

```
(rkllm_env) ubuntu@LAPTOP-CM103FLE:~$ CUDA_VISIBLE_DEVICES=0 ~/rkllm_env/bin/python3 qwen_rkllm_converter.py
INFO: rkllm-toolkit version: 1.3.0
Loading Hugging Face model directory into memory...
[transformers] The fast path is not available because one of the required library is not installed. Falling back to torch implementation. To install follow
https://github.com/Flan-ai-org/flash-linear-attention#installation and https://github.com/Dao-AI-Lab/causal-convid
WARNING: rkllm-toolkit only exports Qwen3_5ForCausalLM of Qwen3_5ForConditionalGeneration!
Building and quantizing the model to w8a8 precision for RK3588...
Optimizing model: 100% | 535/535 [00:29<00:00, 18.18it/s]
Exporting the final deployment binary file...
INFO: Setting eos to 248046
INFO: Setting add_bos_token to False
Converting model: 100% | 427/427 [00:00<00:00, 2480564.83it/s]
INFO: Setting max_context_limit to 4096
INFO: Exporting the model, please wait ...
[=====] 609/609 (100%)
INFO: Model saved to ./qwen3_5_4b.rkllm!
Success! Your deployment asset 'qwen3_5_4b.rkllm' is ready for your board.
```

```
qwen3_5_4b.rkllm
(rkllm_env) ubuntu@LAPTOP-CM103FLE:~$ ls -las qwen3_5_4b.rkllm
5411084 -rw-r--r-- 1 ubuntu ubuntu 5540941884 Aug  1 08:20 qwen3_5_4b.rkllm
```

Next to step to use: rkllm on rk3588 chip Refer this link

https://docs.google.com/document/d/1q6IRA5ijqBpLqVsv6u-jv5sdXb061_dQB-aa8NIP0Y4/e/dit?usp=drive_link